# Science Curriculum

Physics, biology, and chemistry reference material for introductory lessons.

Reference material for lesson planning and course-content development.

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# 1. Teaching science from observation to explanation

Science lessons should distinguish what is observed from how the observation is explained. Learners often jump directly to a memorized explanation. A stronger routine is to ask what happened, what evidence supports that description, and then which scientific idea explains it.

Models are simplified representations. A cell diagram, force arrow, or chemical equation leaves out many details on purpose. Teachers should say what the model is useful for and avoid treating it as a literal picture of every part of reality.

- Use units consistently in calculations.
- Label diagrams clearly.
- Ask learners to predict before showing the result of a demonstration.
- Separate a measured result from an interpretation.

Practical activities should use safe, simple materials and should not rely on a learner performing hazardous experiments independently.

## 2. Motion, speed, velocity, and acceleration

Distance is the total path traveled. Displacement describes the change in position from start to finish and includes direction. Speed describes how fast distance changes with time. Velocity includes direction. Acceleration describes how velocity changes.

### Average speed

Average speed is total distance divided by total time. Learners should include units and should recognize that the average does not mean the object traveled at that exact speed at every moment.

### Velocity and direction

If an object moves east and then returns west, its distance and displacement can differ substantially. Round-trip examples are useful because the final displacement can be zero while distance is not.

### Acceleration

Acceleration can occur because speed changes, direction changes, or both. A negative acceleration value depends on the chosen direction and should not automatically be described as 'slowing down'.

- Use motion tables before graph interpretation.
- Ask learners to compare distance-time and velocity-time descriptions.
- Include one question where units reveal an error.

# 3. Forces and Newton's laws

A force is an interaction that can change motion. Learners should reason about net force, not simply count the number of forces. Two equal opposite forces can produce zero net force even though forces are present.

## First law

When net force is zero, an object maintains its current state of motion. An object moving at constant velocity does not need a forward net force to 'keep it moving' in the idealized model.

## Second law

Net force, mass, and acceleration are related. For the same mass, a larger net force produces a larger acceleration. For the same net force, a larger mass produces a smaller acceleration.

## Third law

Interaction forces come in equal and opposite pairs acting on different objects. The fact that the forces are equal does not mean the objects must have equal accelerations, because their masses may differ.

## Free-body reasoning

A free-body diagram should show forces acting on one chosen object. Avoid mixing forces acting on different objects in one diagram. Label direction and type clearly.

# 4. Work, energy, and power

Work is done when a force acts through a displacement in the direction of the force component. Energy is a quantity associated with the ability to cause change. Power describes how quickly work is done or energy is transferred.

## Kinetic energy

Kinetic energy increases with mass and with the square of speed. This means doubling speed has a larger effect than doubling mass. Use numerical comparisons to make the squared relationship visible.

## Gravitational potential energy

Near Earth's surface, gravitational potential energy depends on mass, gravitational field strength, and height relative to a chosen reference. The reference level should be stated when it matters.

## Energy transfer

In real systems, some energy may be transferred to thermal energy, sound, or deformation. Avoid saying energy is 'lost' when the intended idea is that it has moved into less useful forms.

- Include diagrams showing energy before and after a process.
- Ask learners to identify the system and the energy stores involved.

# 5. Cell structure and function

Cells are the basic structural and functional units of living organisms. Introductory lessons should focus on major structures and their roles rather than long lists of organelles.

## Core structures

- Cell membrane: controls movement of substances into and out of the cell.
- Cytoplasm: contains many chemical reactions and cellular components.
- Nucleus: contains most genetic material in eukaryotic cells.
- Mitochondria: involved in cellular respiration and energy transfer.
- Ribosomes: sites of protein synthesis.

## Plant and animal cells

Plant cells typically include a cellulose cell wall, a large permanent vacuole, and chloroplasts in photosynthetic tissues. Both plant and animal cells have cell membranes, cytoplasm, DNA, and ribosomes.

## Common misconception

Mitochondria are often called the 'powerhouse of the cell', but the phrase can become empty memorization. Learners should connect mitochondria to cellular respiration and ATP production rather than repeat the nickname.

# 6. DNA, genes, transcription, and translation

DNA stores genetic information in a sequence of bases. A gene is a region of DNA that can contribute to a functional product, often through RNA and protein production. Keep the central idea clear before adding molecular detail.

## Transcription

During transcription, a DNA template is used to build a complementary RNA molecule. In eukaryotic cells, transcription occurs in the nucleus. The RNA can then be processed before leaving the nucleus.

## Translation

During translation, ribosomes read mRNA codons and use tRNA molecules to bring amino acids. The amino-acid sequence forms a polypeptide that can fold into a functional protein.

## Mutations

A mutation is a change in DNA sequence. Its effect can range from none to significant depending on where the change occurs and how it affects the final product. Avoid saying every mutation is harmful.

- Use a short base-pairing example.
- Separate the location and purpose of transcription from translation.
- Ask learners to trace information flow from DNA to protein.

# 7. Photosynthesis and cellular respiration

Photosynthesis captures light energy and stores it in chemical form. Cellular respiration transfers energy from organic molecules into ATP that cells can use. The two processes are related but should not be taught as simple mirror images.

## Photosynthesis

The overall process uses carbon dioxide and water and produces carbohydrates and oxygen in photosynthetic organisms. Chloroplasts contain the structures involved in light capture and carbon fixation.

## Cellular respiration

Respiration breaks down energy-rich molecules through several stages. In aerobic respiration, oxygen serves as the final electron acceptor. Mitochondria are central to later stages in eukaryotic cells.

## Common misconceptions

- Plants also carry out cellular respiration.
- Photosynthesis does not occur only at night or only during one single reaction.
- Oxygen produced in photosynthesis comes from water, not directly from carbon dioxide.

At an introductory level, focus on inputs, outputs, locations, and energy purpose before biochemical pathway detail.

# 8. Atoms, ions, and isotopes

Atoms contain protons, neutrons, and electrons. The number of protons defines the element. The mass number is the total number of protons and neutrons. Neutral atoms have equal numbers of protons and electrons.

## Ions

Ions form when the number of electrons differs from the number of protons. Losing electrons creates a positive ion; gaining electrons creates a negative ion. The nucleus does not change in an ordinary ion-forming process.

## Isotopes

Isotopes are atoms of the same element with the same number of protons but different numbers of neutrons. They have the same atomic number but different mass numbers.

## Common mistakes

- Changing the number of protons changes the element.
- An ion is not the same thing as an isotope.
- Electrons contribute very little to atomic mass compared with protons and neutrons.

Use small particle-count tables so learners can identify element, mass number, and charge.

# 9. Chemical reactions and balancing equations

Chemical reactions rearrange atoms into new combinations. The atoms are not created or destroyed in ordinary chemical reactions, so equations must conserve the number of atoms of each element.

## Reactants and products

Reactants appear before the reaction arrow and products after it. Learners should connect the symbolic equation to the substances and to particle rearrangement.

## Balancing

Balance by changing coefficients, not the chemical formulas. Changing $H_2O$ to $H_2O_2$ in order to balance oxygen changes the substance and is not a valid balancing step.

## Example

For hydrogen reacting with oxygen to form water, $2H_2 + O_2 \rightarrow 2H_2O$ balances four hydrogen atoms and two oxygen atoms on both sides.

- Start with equations containing two or three substances.
- Count atoms explicitly before introducing shortcuts.
- Include one incorrect attempt where a subscript was changed.

# 10. Acids, bases, and pH

Acid-base lessons should connect the pH scale to observable chemical behavior while keeping safety clear. Introductory learners do not need a complete treatment of every acid-base theory.

## pH scale

At room temperature, pH values below 7 are acidic, 7 is neutral, and values above 7 are basic in dilute aqueous solutions. The pH scale is logarithmic, so a one-unit change represents a tenfold change in hydrogen-ion concentration.

## Indicators

Indicators change color over a pH range. They can show whether a solution is acidic or basic but may not provide an exact pH. A digital pH meter provides a more direct measurement.

## Neutralization

Acids and bases can react to form products that may include a salt and water. The exact products depend on the reactants.

- Do not encourage unsupervised mixing of household chemicals.
- Use pre-defined safe classroom examples or supplied data tables.
- Ask learners to interpret pH values rather than memorize a list of substances.